

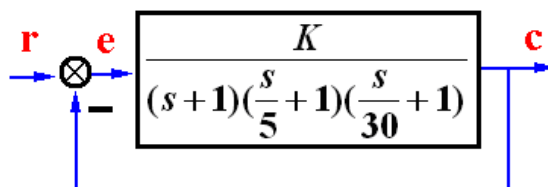
串联PID校正

例1 已知单位反馈系统的开环传递函数：

$$G(s) = \frac{K}{(s+1)(\frac{s}{5}+1)(\frac{s}{30}+1)}$$

设计PID控制器，要求：

$$\begin{cases} r(t) = t \rightarrow e_{ss}^* \leq 0.1 \\ \sigma \% \leq 20 \% \\ t_s \leq 0.5 \end{cases}$$



试确定 $G_c(s)$ 。

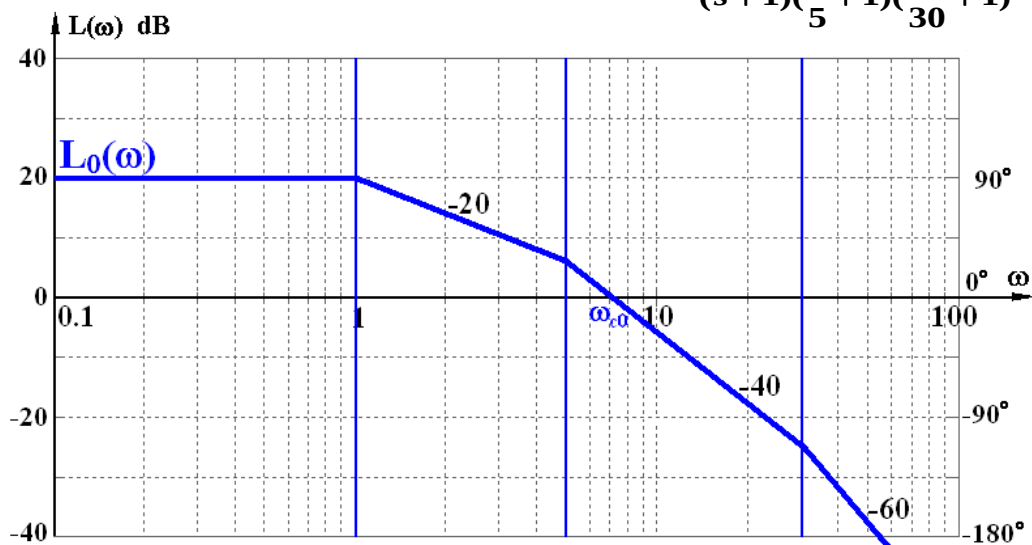
解. (1) 由要求：

$$\begin{cases} r(t) = t \rightarrow e_{ss}^* \leq 0.1 \\ \sigma \% \leq 20 \% \\ t_s \leq 0.5 \end{cases}$$

查 P173 图 5-56 $\Rightarrow$

$$\begin{cases} K = 10 \quad (v=1) \\ \gamma^* \geq 65^\circ \\ t_s = 6.8/\omega_c^* \leq 0.5 \rightarrow \omega_c^* \geq 6.8/0.5 = 13.6 \end{cases}$$

(2) 绘制 $L_0(\omega)$ 曲线如图所示：

$$G(s) = \frac{K}{(s+1)(\frac{s}{5}+1)(\frac{s}{30}+1)}$$


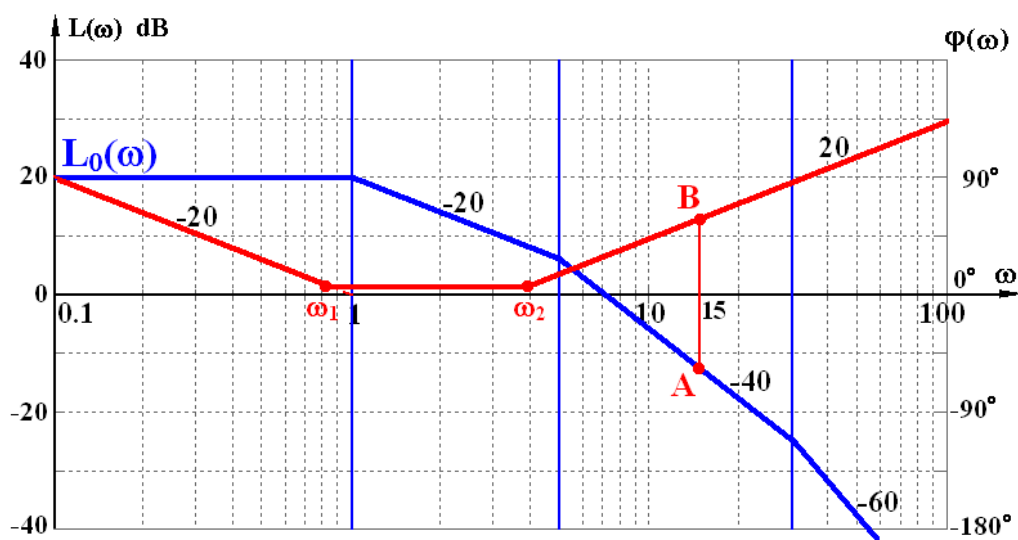
$$\omega_{c0} = \sqrt{5 \times 10} = 7.07 < 13.6 = \omega_c^*$$

$$\begin{aligned}\gamma_0 &= 180^\circ - \arctan 7.07 - \arctan \frac{7.07}{5} - \arctan \frac{7.07}{30} \\ &= 180^\circ - 81.95^\circ - 54.74^\circ - 13.26^\circ = 30^\circ < 65^\circ = \gamma^*\end{aligned}$$

$$\begin{aligned}\gamma_0(\omega_c^*) &= 180^\circ - \arctan 13.6 - \arctan \frac{13.6}{5} - \arctan \frac{13.6}{30} \\ &= 180^\circ - 85.79^\circ - 69.81^\circ - 24.39^\circ = 4.789^\circ < 65^\circ = \gamma^*\end{aligned}$$

③ 确定校正形式：

选择 $\omega_c = 15 > 13.6$, $\rightarrow$ A, B

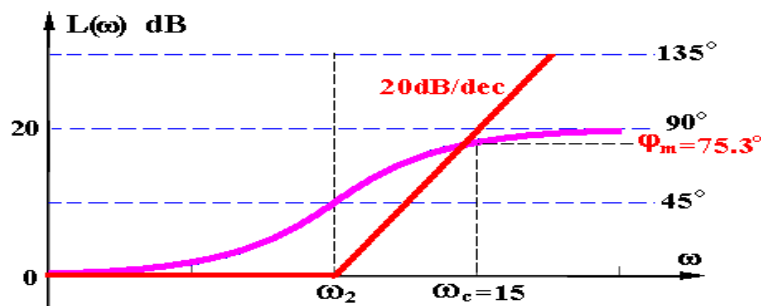


$$\varphi_m = \gamma^* - \gamma_0(\omega_c) + 6^\circ$$

$$\begin{aligned}\gamma_0(15) &= 180^\circ - \arctan 15 - \arctan \frac{15}{5} - \arctan \frac{15}{30} \\ &= 180^\circ - 86.2^\circ - 71.6^\circ - 26.5^\circ = -4.3^\circ\end{aligned}$$

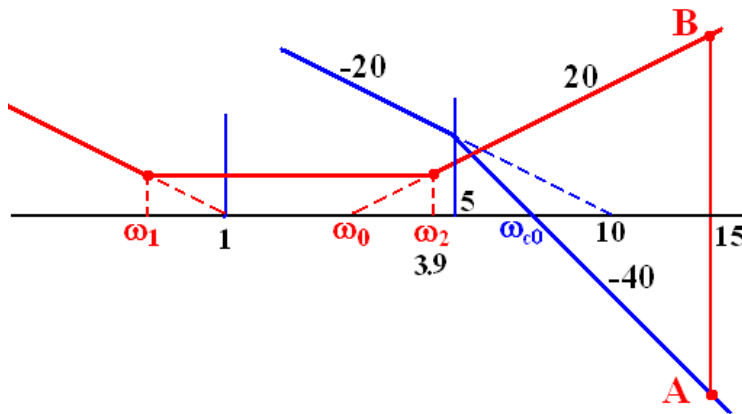
$$\varphi_m = 65^\circ + 4.3^\circ + 6^\circ = 75.3^\circ$$

设第二个转折频率为 ω_2 , 则有



$$\angle \left(1 + \frac{s}{\omega_2} \right)_{s=j15} = \arctan \frac{15}{\omega_2} = \varphi_m = 75.3^\circ$$

$$\omega_2 = \frac{15}{\tan 75.3^\circ} = 3.9$$



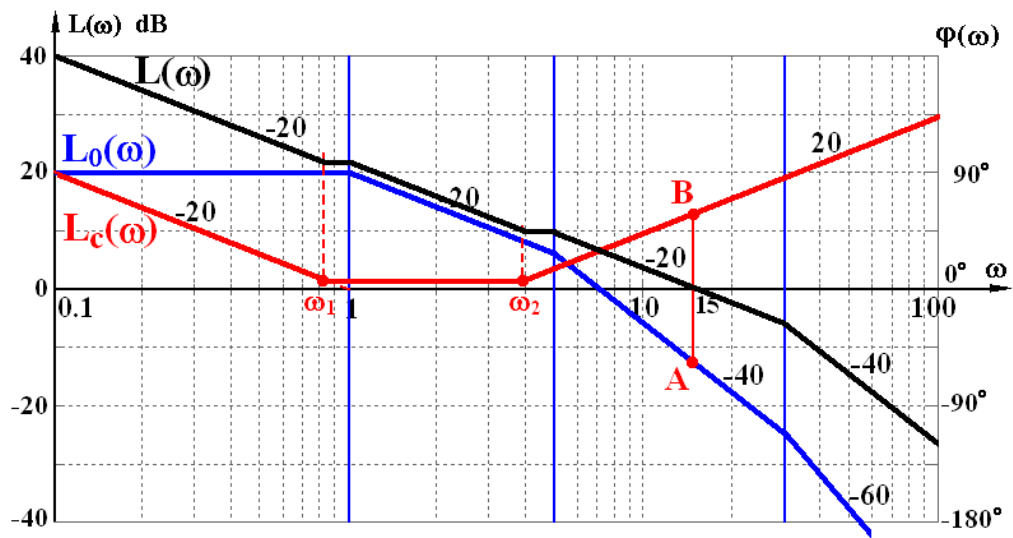
$$\frac{15}{\omega_{c0}} = \frac{\omega_{c0}}{\omega_0} \rightarrow \omega_0 = \frac{\omega_{c0}^2}{15} = \frac{50}{15} = \frac{10}{3}$$

$$\omega_2 = \frac{15}{\tan 75.3^\circ} = 3.9$$

$$\frac{\omega_2}{\omega_0} = \frac{1}{\omega_1} \rightarrow \omega_1 = \frac{\omega_0}{\omega_2} = \frac{10}{3 \times 3.9} = 0.855$$

$$\begin{aligned} G_c(s) &= \frac{\left(\frac{s}{\omega_1} + 1\right)\left(\frac{s}{\omega_2} + 1\right)}{s} = \frac{\left(\frac{s}{0.855} + 1\right)\left(\frac{s}{3.9} + 1\right)}{s} \\ &= \frac{0.3s^2 + 1.426s + 1}{s} = \frac{K_D s^2 + K_P s + K_I}{s} \end{aligned}$$

$$\begin{cases} K_D = 0.3 \\ K_P = 1.426 \\ K_I = 1 \end{cases}$$



⑤ 验算：

$$G(s) = G_c(s) \cdot G_0(s) = \frac{10\left(\frac{s}{0.855} + 1\right)\left(\frac{s}{3.9} + 1\right)}{s(s+1)\left(\frac{s}{5} + 1\right)\left(\frac{s}{30} + 1\right)}$$

$$\omega_c = 15 \geq 13.6 = \omega_c^*$$

$$\begin{aligned} \gamma &= 180^\circ + \arctan \frac{15}{0.855} + \arctan \frac{15}{3.9} \\ &\quad - 90^\circ - \arctan 15 - \arctan \frac{15}{5} - \arctan \frac{15}{30} \\ &= 180^\circ + 86.7^\circ + 74.5^\circ - 90^\circ - 86.2^\circ - 71.6^\circ - 26.5^\circ \\ &= 67.85^\circ > 65^\circ = \gamma^* \end{aligned}$$

